

COURSE-END PROJECT

Analyzing Customer Orders Using Python

1. Project Overview

This project analyzes customer orders using Python data structures and control structures. The objective is to classify customers according to their total spending, organize products by category, identify purchasing patterns, and generate useful business insights for an e-commerce company.

2. Objectives

- Store customer order information using lists, tuples, and dictionaries.
- Classify products into their respective categories and identify unique categories.
- Calculate total spending for each customer and classify customers as high-value, moderate, or low-value buyers.
- Calculate category-wise sales and identify important purchasing insights.
- Identify unique products, electronics customers, top-spending customers, and customers purchasing across multiple categories.
- Generate a final business summary that can support marketing and inventory decisions.

3. Data Structures and Python Concepts Used

Python Concept	Purpose
List	Store customer names and collections of records
Tuple	Store each order as customer, product, price, and category
Dictionary	Map customers to products and products to categories
Set	Find unique categories/products and perform set operations
Loops	Calculate spending and process order information
Conditionals	Classify customers based on spending
List Comprehension	Find customers who purchased electronics
Sorting	Identify the top three highest-spending customers

4. Customer Classification

Customers were classified using their total purchase value. A customer spending above \$100 was classified as a high-value buyer, a customer spending between \$50 and \$100 as a moderate buyer, and a customer spending below \$50 as a low-value buyer.

Customer	Classification
Ram	High-value buyer
Sita	High-value buyer
Aman	Low-value buyer
Rahul	Low-value buyer

Priya	High-value buyer
Vikas	High-value buyer
Neha	High-value buyer
Rohit	Moderate buyer
Pooja	Moderate buyer
Arjun	Moderate buyer

Customer Group	Number of Customers
High-value	5
Moderate	3
Low-value	2

5. Category-wise Sales Analysis

The total sales generated by each product category were calculated and stored in a dictionary.

Category	Total Sales
Electronics	\$2,440
Clothing	\$285
Home Essentials	\$290

6. Key Business Insights

- Electronics is the highest-revenue category, with total sales of \$2,440.
- Ram is the highest-spending customer, with total spending of \$1,280.
- There are 5 high-value customers, 3 moderate customers, and 2 low-value customers.
- The most frequently purchased products identified in the analysis are Shoes and Mobile, with 2 purchases each.
- Category-wise sales indicate that Electronics is the strongest-performing category in the dataset.

7. Additional Analysis Performed

The project also extracted unique products using a set, identified customers who purchased electronics using list comprehension, found the top three highest-spending customers through sorting, and used set operations to identify customers purchasing from multiple categories and customers common to Electronics and Clothing. Dictionary methods were also used to retrieve and modify customer records.

8. Conclusion

The project demonstrates how Python's built-in data structures and control structures can be applied to a real-world e-commerce order analysis problem. The analysis identifies high-value customers, popular products, and category-level sales performance. These insights can help a business focus marketing efforts on valuable customers and plan inventory around stronger-performing product categories.

9. Final Business Result

Metric	Result
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Highest Revenue Category	Electronics
Highest Spending Customer	Ram — \$1,280
High-Value Customers	5
Moderate Customers	3
Low-Value Customers	2